

VALLURI DINESH RAM

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AI/ML Engineer | Agentic Systems | Software Engineer | Portfolio: dineshramv13.github.io

SKILLS

Languages & Databases : Python, C, HTML, CSS, SQL, JavaScript, FAISS, ChromaDB, Pinecone, Vector Database
Frameworks & Tools : Scikit-Learn, TensorFlow, Keras, **LangChain**, LangGraph, Streamlit, **LLM**, OpenCV, **Generative AI**, Agentic AI, Prompt Engineering, **Machine Learning**, Deep Learning, DSA
Cloud & Workflow : AWS, GitHub, **Git**, MLOps, **Django**, Flask, FastAPI, CI/CD, GitHub Actions, Docker

EXPERIENCE

Junior Software Engineer - AI/ML, Tractrix Opto Dynamics ([website](#)) **Feb 2025 - August 2026**

- Architected system design for a self-reflective RAG (Retrieval-Augmented Generation) system as an orchestration workflow of an agentic, stateful LangGraph pipeline with adaptive retrieval gating, LLM-based relevance scoring, and multi-stage conditional routing, improving grounded response accuracy from 75% to 90% over 500 pages of technical documentation through automated hallucination detection and autonomous answer revision.
- Optimized a real-time GStreamer video pipeline on embedded Linux devices, reducing end-to-end inference latency by 35% and increasing throughput from 18 FPS to 30 FPS while lowering CPU utilization by 40%. Shipped modular backend services and REST APIs using FastAPI and Flask for camera ingestion, FFmpeg recording pipelines, video transfer, MediaMTX client-server architecture, data annotations, database, stabilization systems across edge platforms.

Artificial Intelligence Intern, Smart Prepper ([website](#)) **Aug 2024 - Nov 2024**

- Developed an end-to-end Flashcard Generator that ingests text, topic, and multi-page PDF input (AWS S3-backed) and produces Q&A flashcards using LLM prompting, token-aware context-window chunking, and RAG-based retrieval automating over 1,000 flashcards per hour. Data visualization using Matplotlib and unified secure PDF upload and multi-page text extraction workflow on AWS S3, with usage analytics showing an 80% successful conversion rate.
- Built a Mind Map Generator (Django and OpenAI) that converts long, unstructured notes into structured Markdown mind maps; cutting manual summarization time by 70% in user trials with UPSC aspirants.
- Implemented a scalable pipeline using vector embeddings and Pinecone for semantic retrieval to support quiz and MCQ generation, improving MCQ relevance by 25% in internal testing and user evaluations.

Technology Lead, GDSC (Google Developer Students Club) Leadership at College **Aug 2023 - Aug 2024**

- As the Lead, conducted sessions on machine learning, deep learning using TensorFlow, Google Colab, Cloud Study Jams. At NLP program, constructed a speaker recognition system with 97% accuracy using CNN architecture. ([Link](#))

PROJECTS

MeetingMind: AI-Powered Meeting Orchestration Agent ([GitHub](#))

- Crafted a multi-agent LangGraph pipeline consisting of (Understanding, Extraction, Reflection, Routing, Execution) using async Python and Pydantic-validated function calling with Human-in-the-Loop (HITL).
- Integrated four OAuth2-authenticated MCP (Model Context Protocol) servers for autonomous task execution and observability, validated across 50 real transcripts, on a FastAPI/SQLAlchemy backend with a Streamlit dashboard.

Advanced Multi-Agent AI Content Writer ([GitHub](#))

- Designed a AI-powered five-agent stateful LangGraph agent orchestration pipeline (Router, Research, Orchestrator, Parallel workers, Reducer) with a classifier routing requests into three modes (closed/hybrid/open).
- Connected Tavily tool and LLM API integration for real-time, citation-grounded research with LangSmith observability; deployed to AWS EC2 using Docker, containerization and GitHub Actions CI/CD pipeline for deployments.

Agentic Healthcare Voice Reception Agent ([GitHub](#))

- Created a voice/text intake agent with safety guardrails, an intent router, and a self-corrective verifier constructed on a hybrid RAG pipeline (BM25, dense vectors, RRF, cross-encoder reranking, HyDE).
- Designed a Whisper STT/TTS pipeline on FastAPI with a custom RAGAS evaluation harness, validating the application's response accuracy and grounding during testing.

EDUCATION

Bachelor of Technology, Computer Science and Engineering at HITAM, CGPA: **9.01**

Nov 2021 - June 2025

ACHIEVEMENTS & CERTIFICATIONS

Published a research paper: ([Link](#)); Participant in the MIT iQuHACK 2023: Microsoft challenge, Amazon ML Challenge (103/5000 teams), National Hackathons, Certifications: NPTEL Elite, Cisco CCNA, IBM AI/ML, Udemy AI/ML.